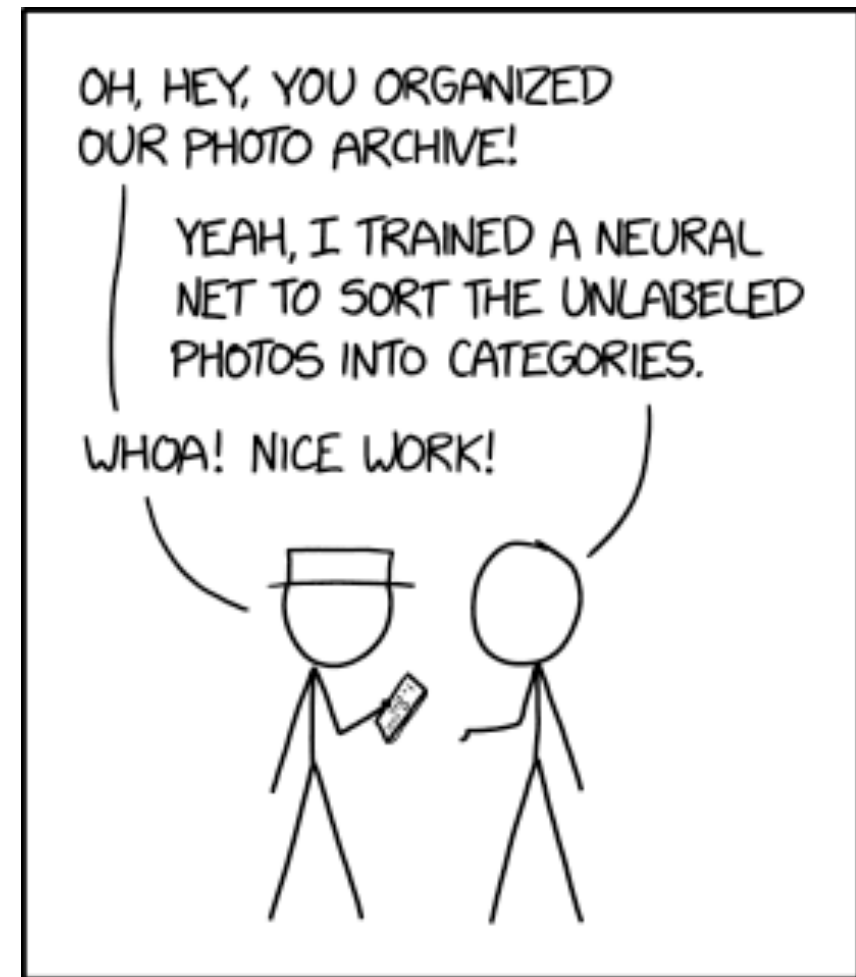


Trustworthy AI

Robert Pless & Zoe Szajnfarber
George Washington University



ENGINEERING TIP:
WHEN YOU DO A TASK BY HAND,
YOU CAN TECHNICALLY SAY YOU
TRAINED A NEURAL NET TO DO IT.

Structure of the class

- Most days will be guest lecture (first hour) + activity (second half)
- Guest lecture is likely to include assigned reading.
- Grades will be based on:
 - Question/comments submitted about assigned reading before lecture via Google form.
 - 3 small projects turned in as a blog post.
 - Final project to be presented as a poster session.
 - Short answer exam based on guest lectures.

Rest of Today:

- My perspective on AI/Machine Learning
- Overview AI/ML/Deep Learning/Reinforcement Learning
- Neural Networks and what makes Deep Learning Deep
- Hint at what is “Trustworthy AI”

Computer Vision

- Originally a mix of Geometry, Statistics and Optics
- Currently nearly completely focused on Machine Learning

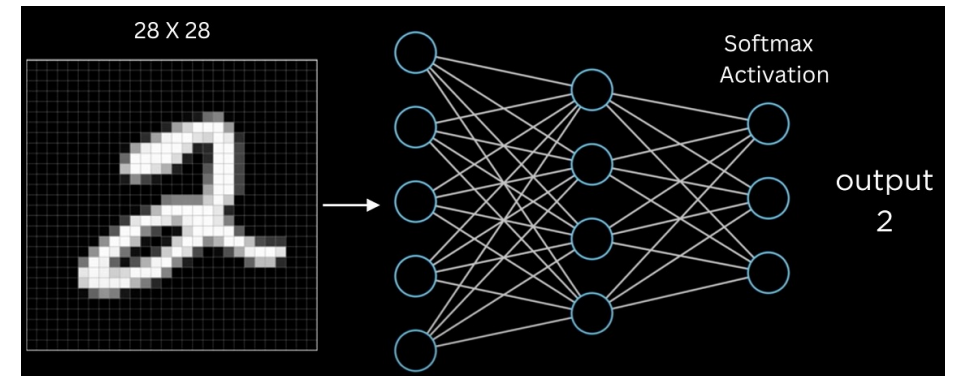


IN CS, IT CAN BE HARD TO EXPLAIN THE DIFFERENCE BETWEEN THE EASY AND THE VIRTUALLY IMPOSSIBLE.

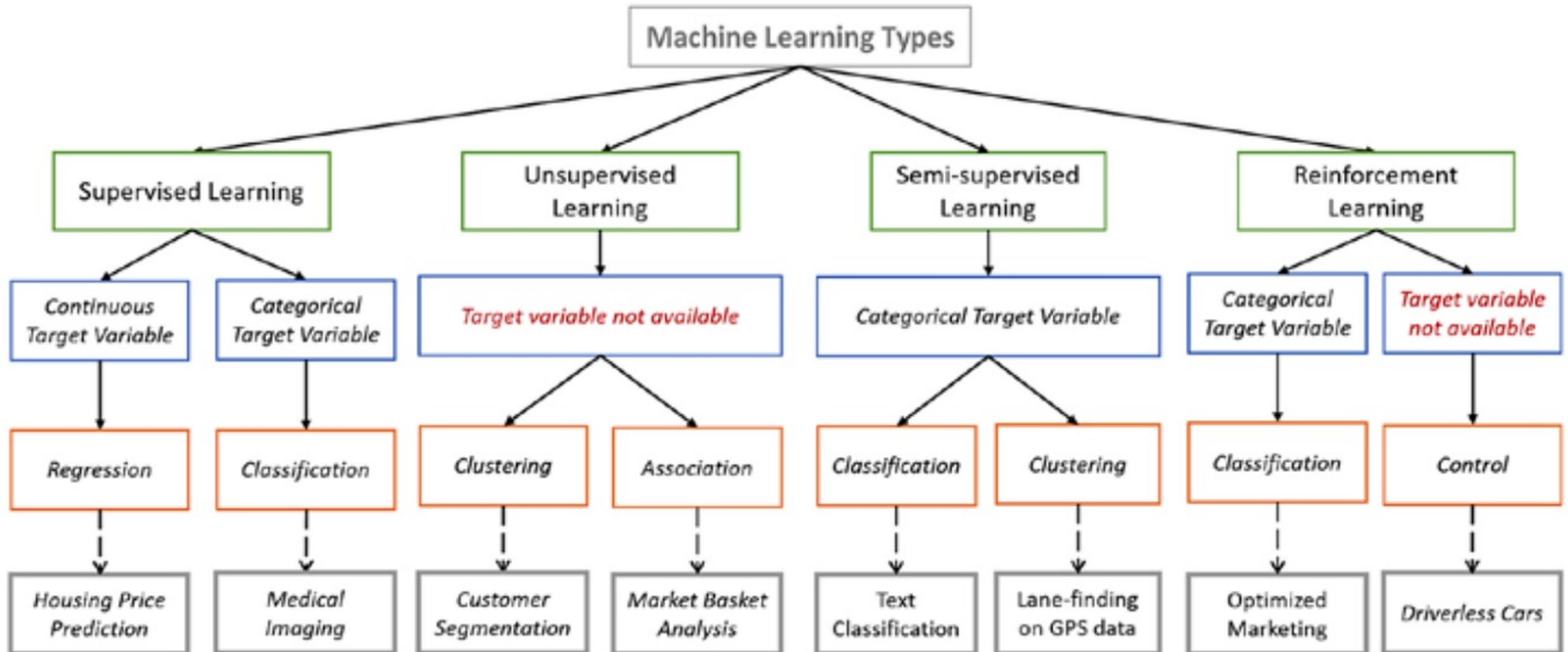
AI vs. Machine Learning

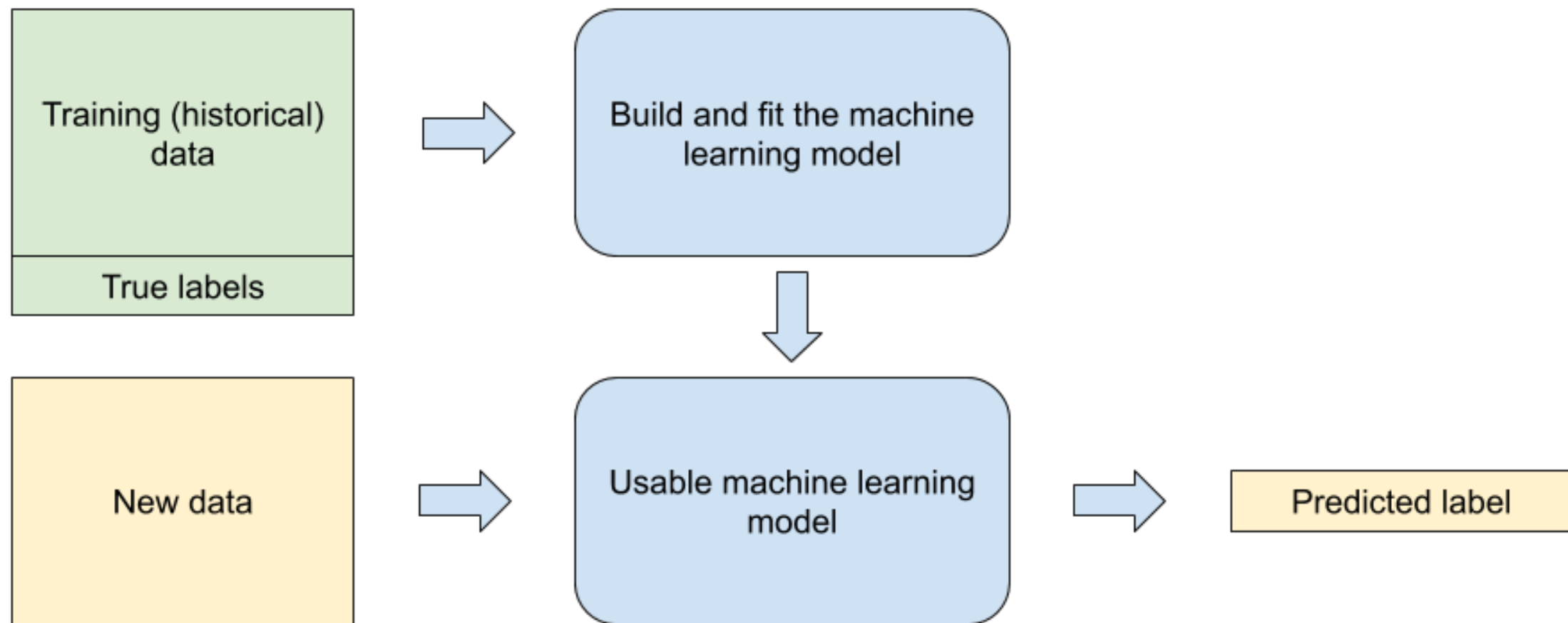
Artificial Intelligence: Systems that can mimic or simulate human-like intelligence (perception, reasoning, decision-making).

Machine Learning: A subset of AI that focuses on algorithms and statistical models to learn patterns from data.



Very broad types of machine learning



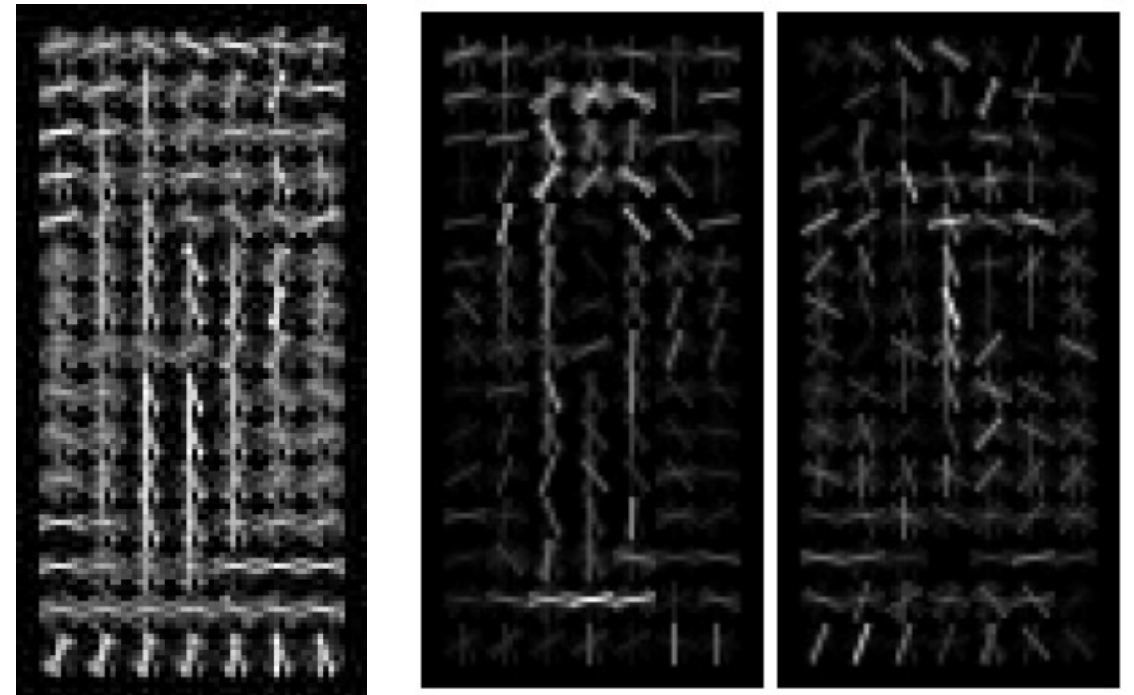


Classical Vs. Modern Machine Learning

Classic approach is to “think really hard” about what features to extract, then use tools like logistic regression, support-vector machines or K-nearest neighbors.

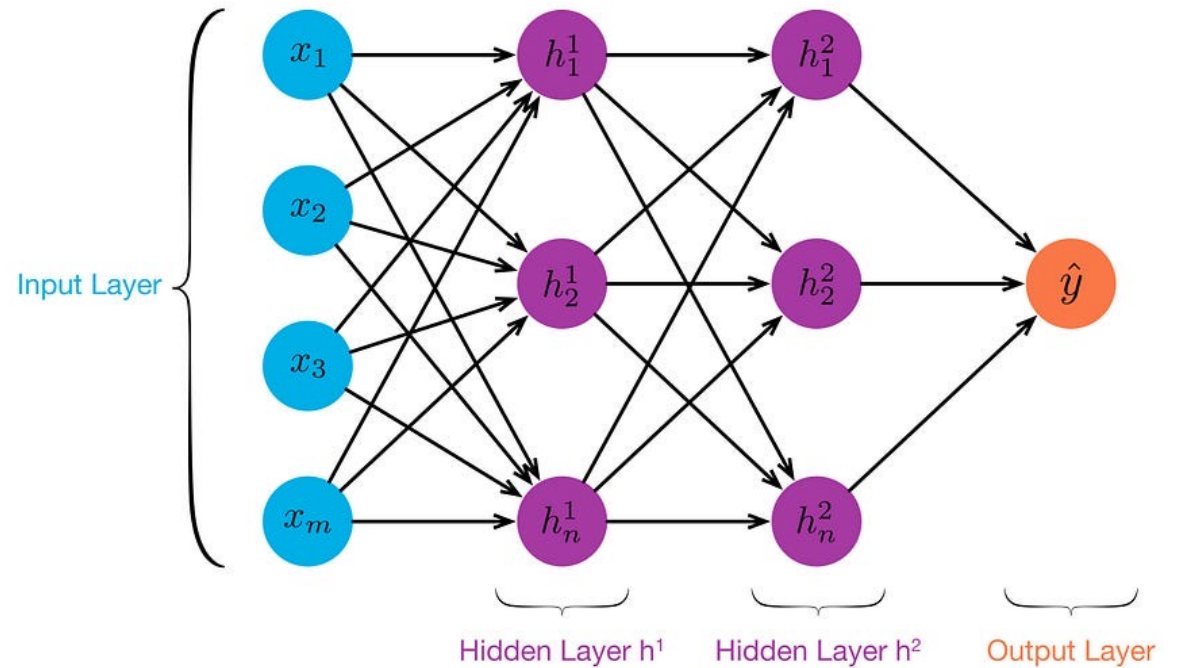
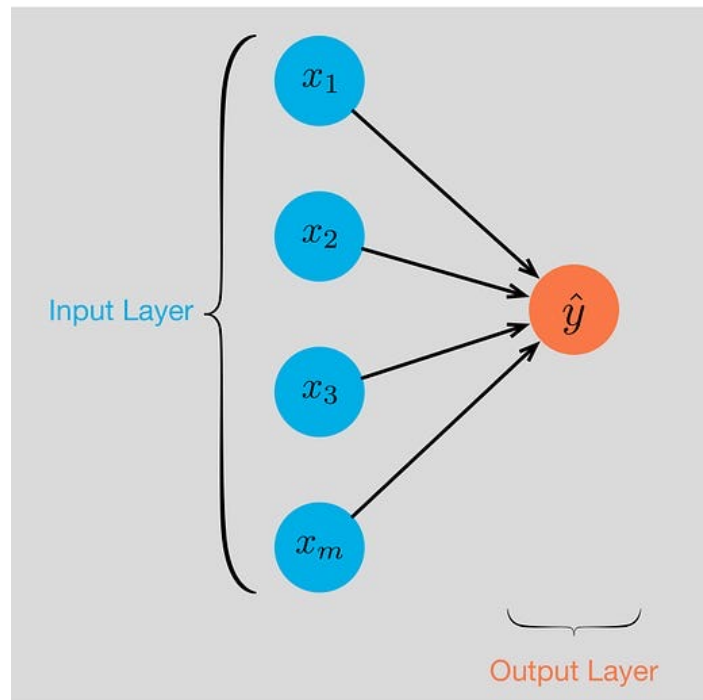


Need ~10,000 example images with labels...
Thousands of parameters



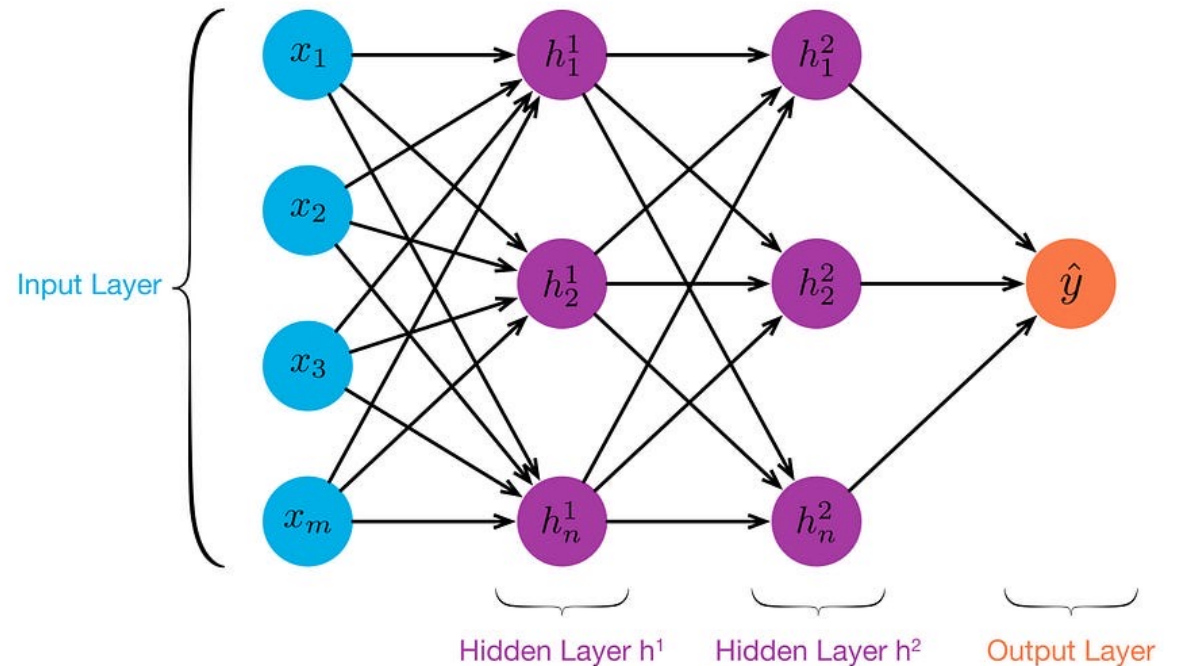
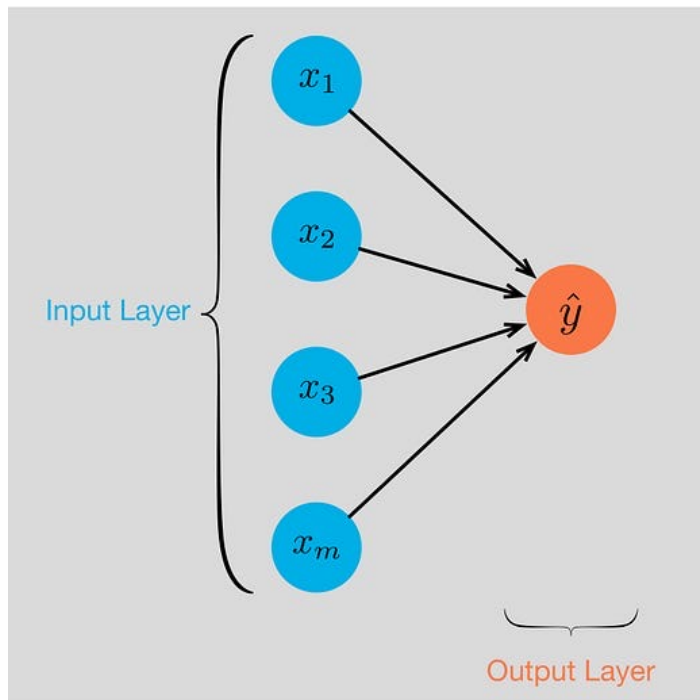
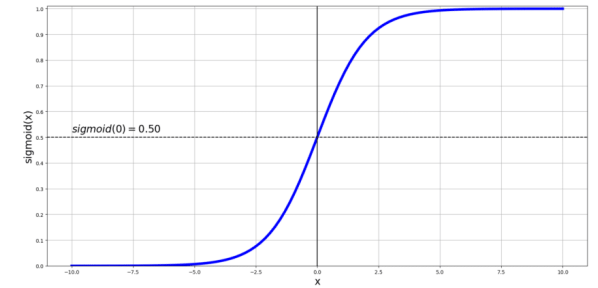
Neural Networks

Neural networks learn their own features; usually linear combinations of input features, and linear (*) combinations of those features.



Neural Networks

Neural networks learn their own features; usually linear combinations of input features, and linear (*) combinations of those features.

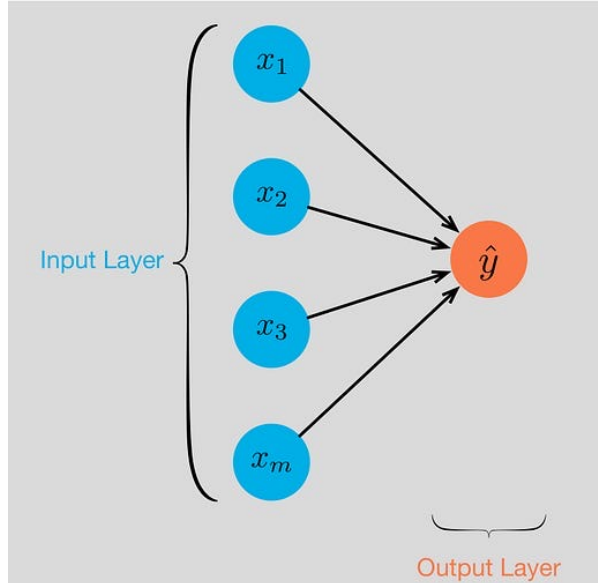


Neural Networks were hot topics in research in Computer Science in 1980's, but stopped for both a theoretical and practical reason

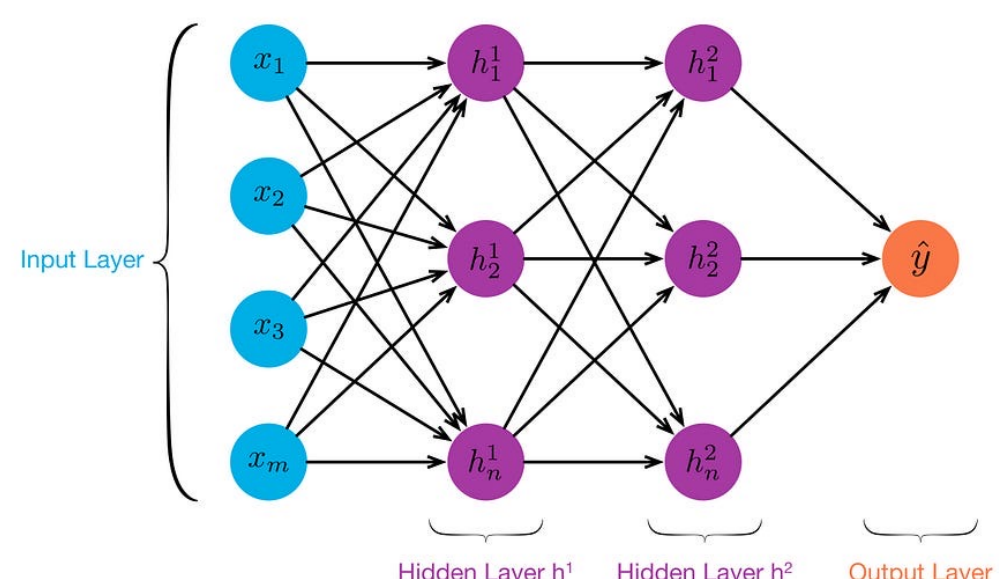
What makes modern models “Deep”?

<https://cs.stanford.edu/people/karpathy/convnetjs/demo/classify2d.html>

(or search for: classify2d karpathy)



$$\hat{y} = f_{\theta}(\mathbf{x}) = \sigma \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right)$$



$$\hat{y} = f_{\theta}(\mathbf{x}) = \sigma \left(\mathbf{W}^{(3)} \sigma \left(\mathbf{W}^{(2)} \sigma \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right) + \mathbf{b}^{(2)} \right) + \mathbf{b}^{(3)} \right)$$

$\theta = (\mathbf{W}, \mathbf{b})$ represents the parameters of the model, consisting of the weight matrix $\mathbf{W}$ and bias vector $\mathbf{b}$.

$$L(\theta) = \frac{1}{2} \|\hat{y} - y\|^2$$

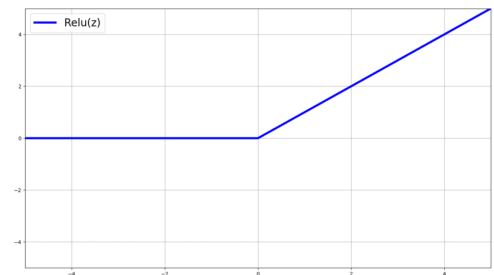
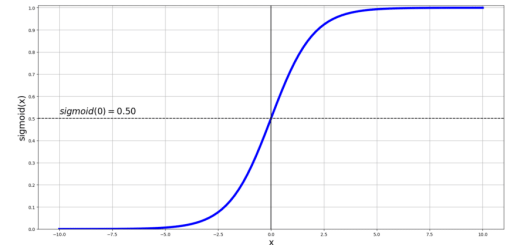
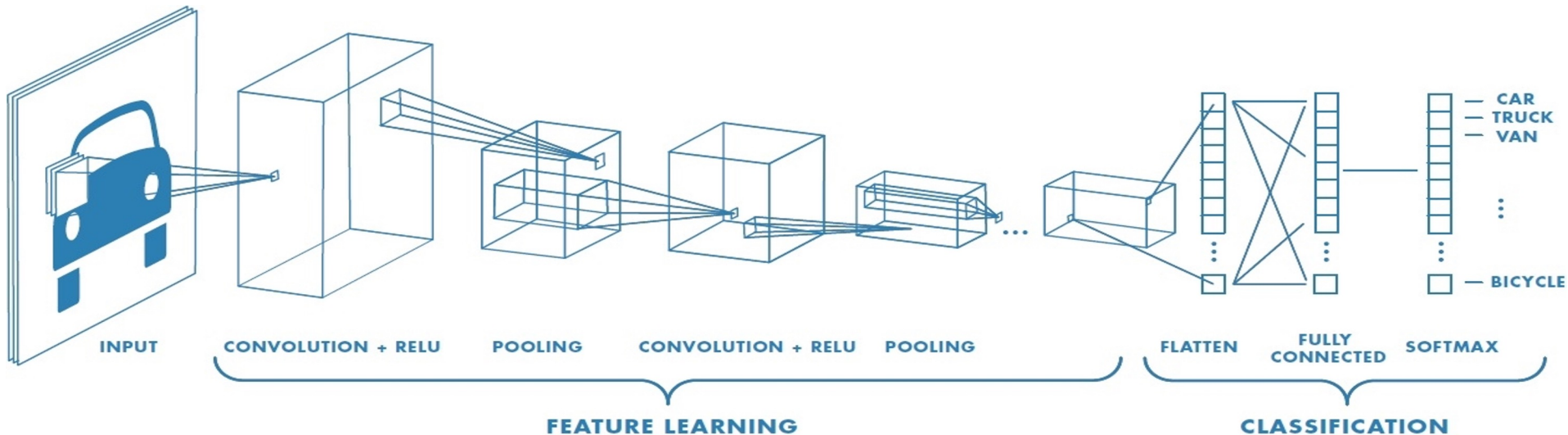
Given example (x, y) , a loss function measures the error of the predicted output $\hat{y}$ compared to y .

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$$

Training is done by computing the gradient of the Loss w.r.t. weights and biases, and updating the parameters, and iterating (many) times. Backpropagation is the name of an efficient algorithm for all the bookkeeping needed to do this when you have this functional form.

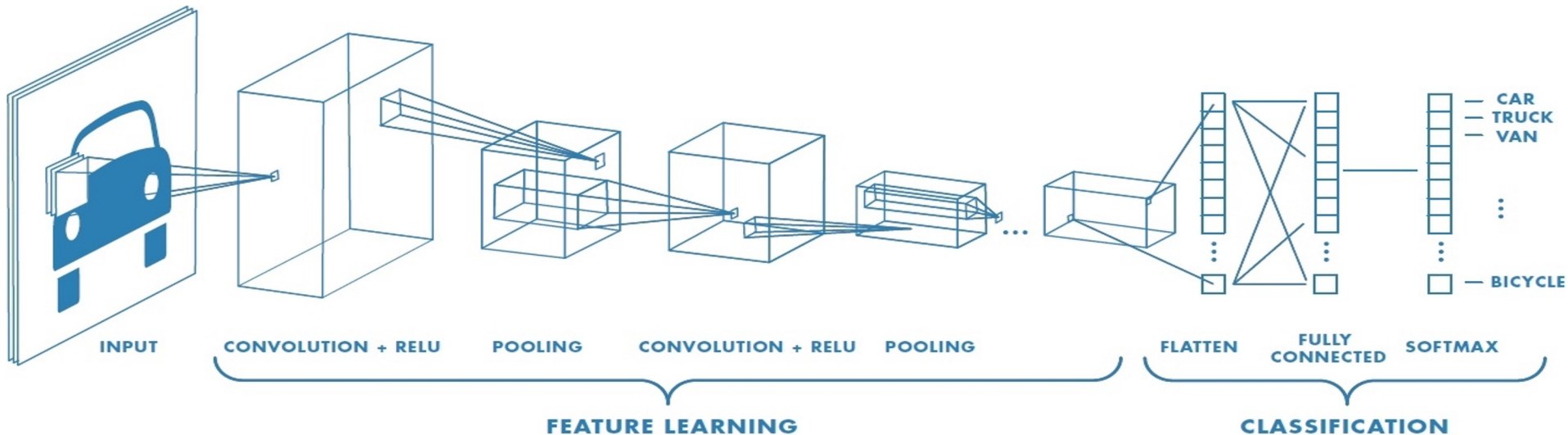
All modern Deep Learning/ Generative AI models are “just” very large neural networks

- Sometimes structured to take advantage of redundancy or regularities in the data...
- Modern networks have hundreds or thousands of layers.
- Modern networks have billions or trillions (!) of weights.

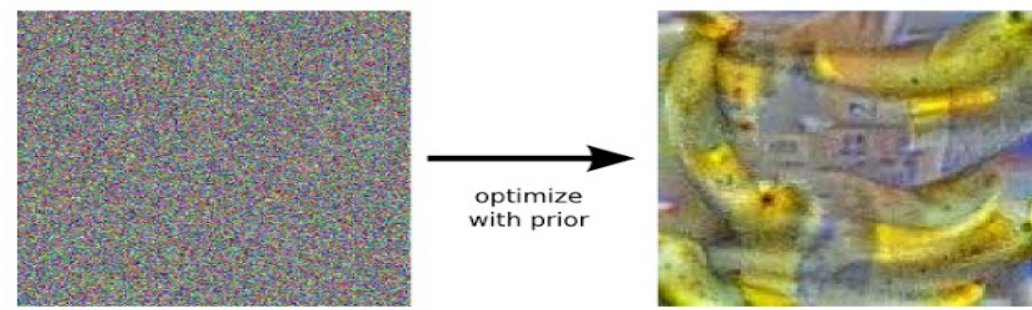


~All modern Deep Learning/ Generative AI models are just **very** large neural networks

- Trained based on very large training datasets. Loss function is the sum of errors over all training examples.
- Backpropagation “needs to keep all examples in memory” to be efficient, so training is broken up into batches that fit into memory at once (and expensive GPUs are useful!). “Stochastic gradient descent”



Network Visualization

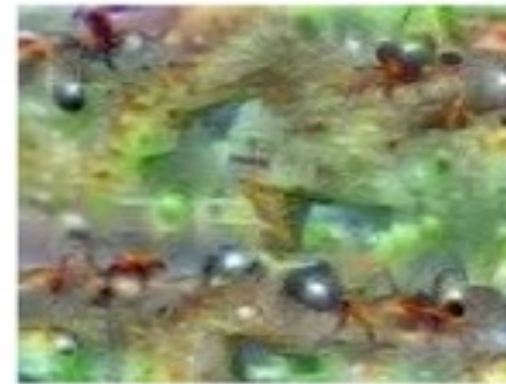


- Instead of training the network to give the right answer, change the “image” to maximize how much it looks like some class A

$$X' = X + \frac{\partial(F(X, W) == \text{class}(A))}{\partial X}$$



Measuring Cup



Ant



Starfish



Banana



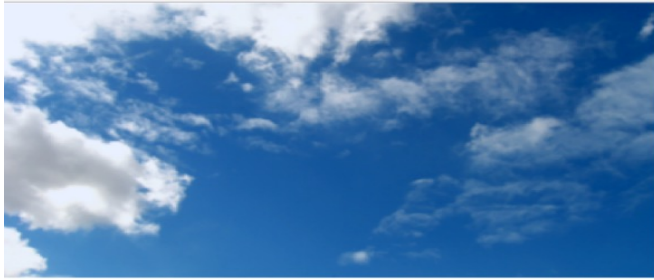
Parachute



Screw

Network Visualiza

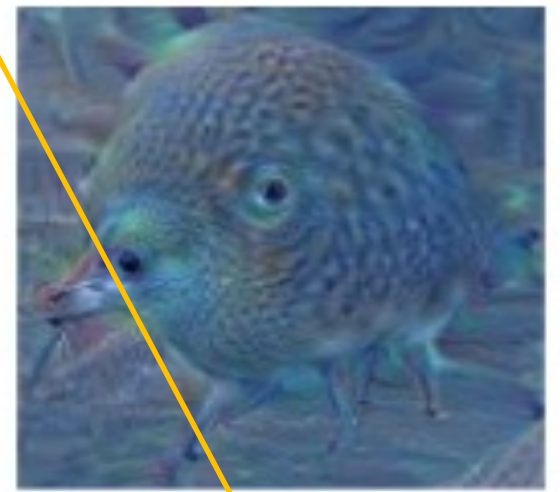
- Google Deep Dream



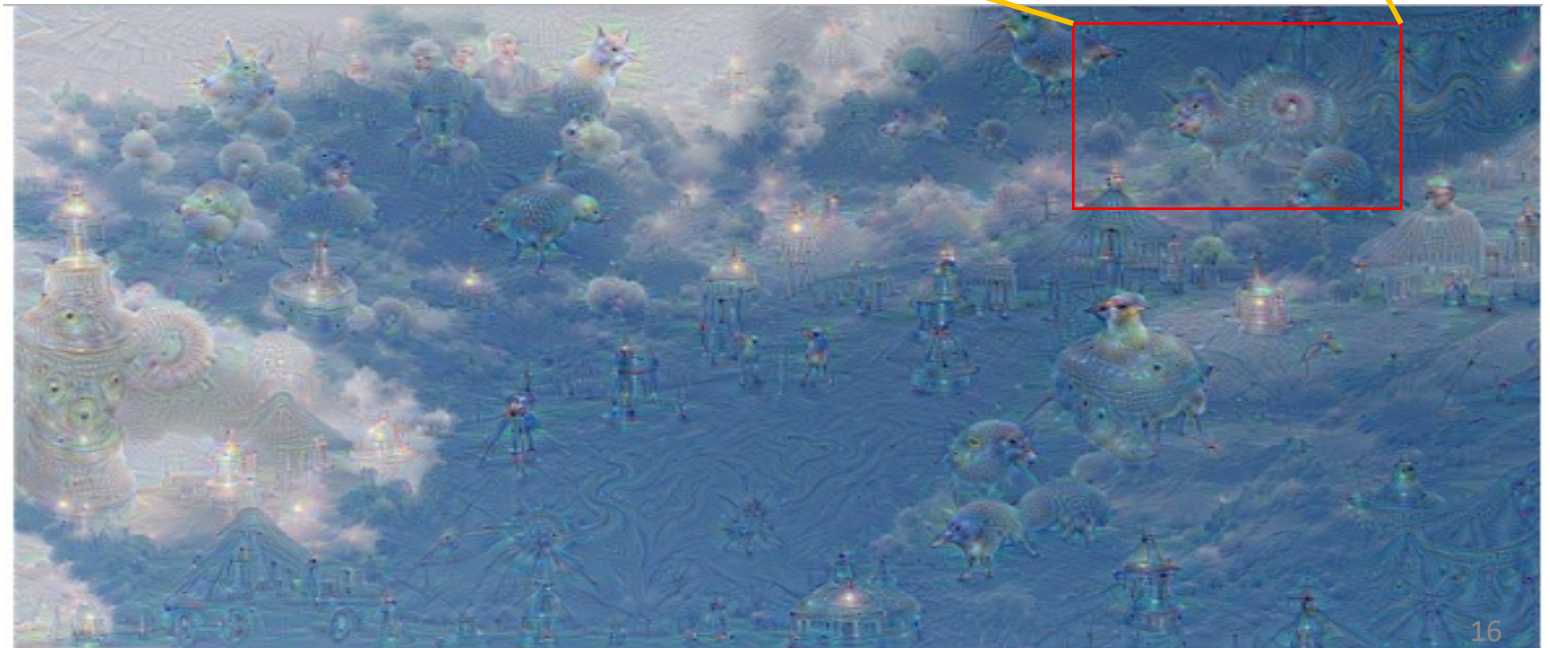
In rectangles in the input image, pick the 2 most active classes, and optimize the image to emphasize those



"The Pig-Snail"



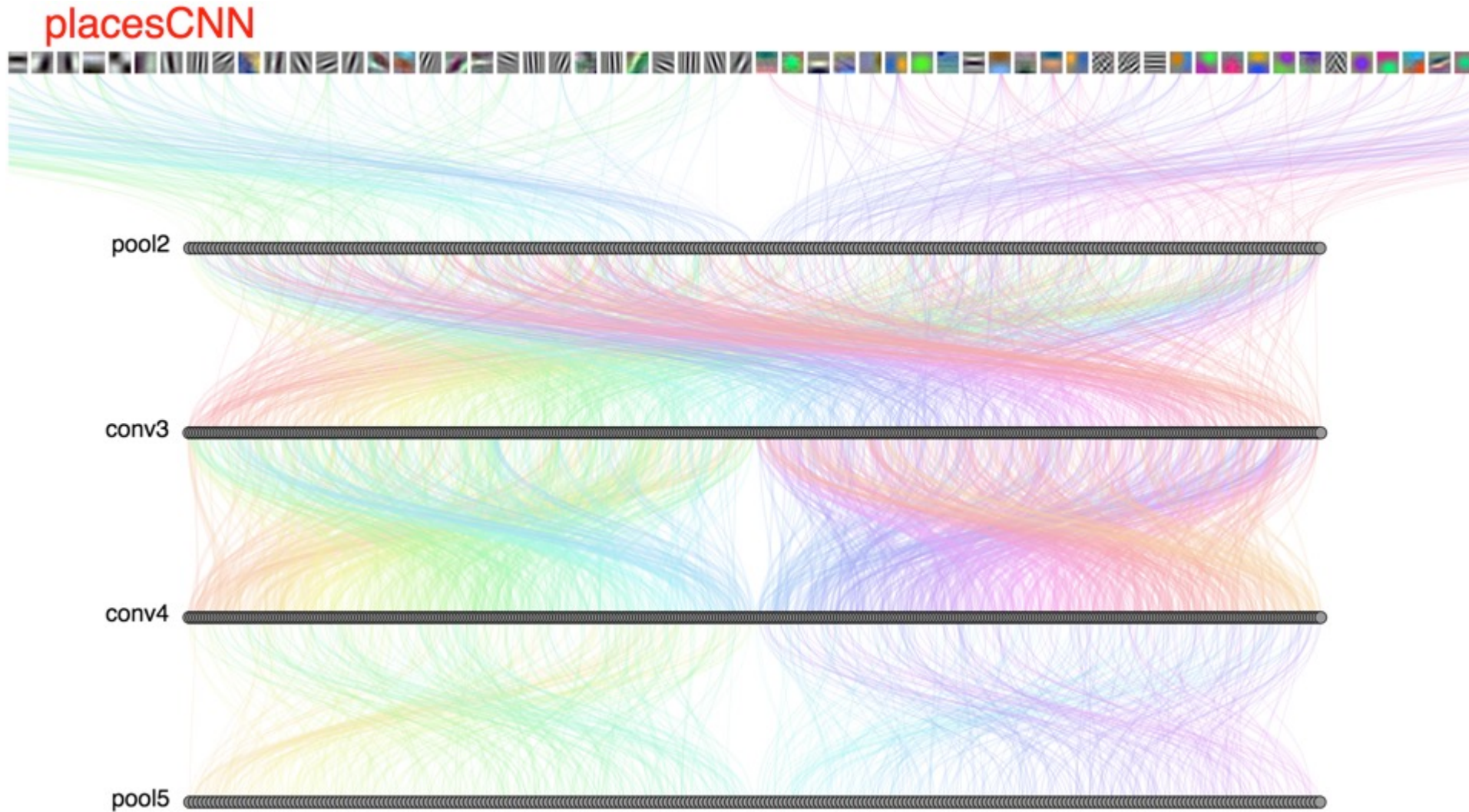
"The Camel-Bird"



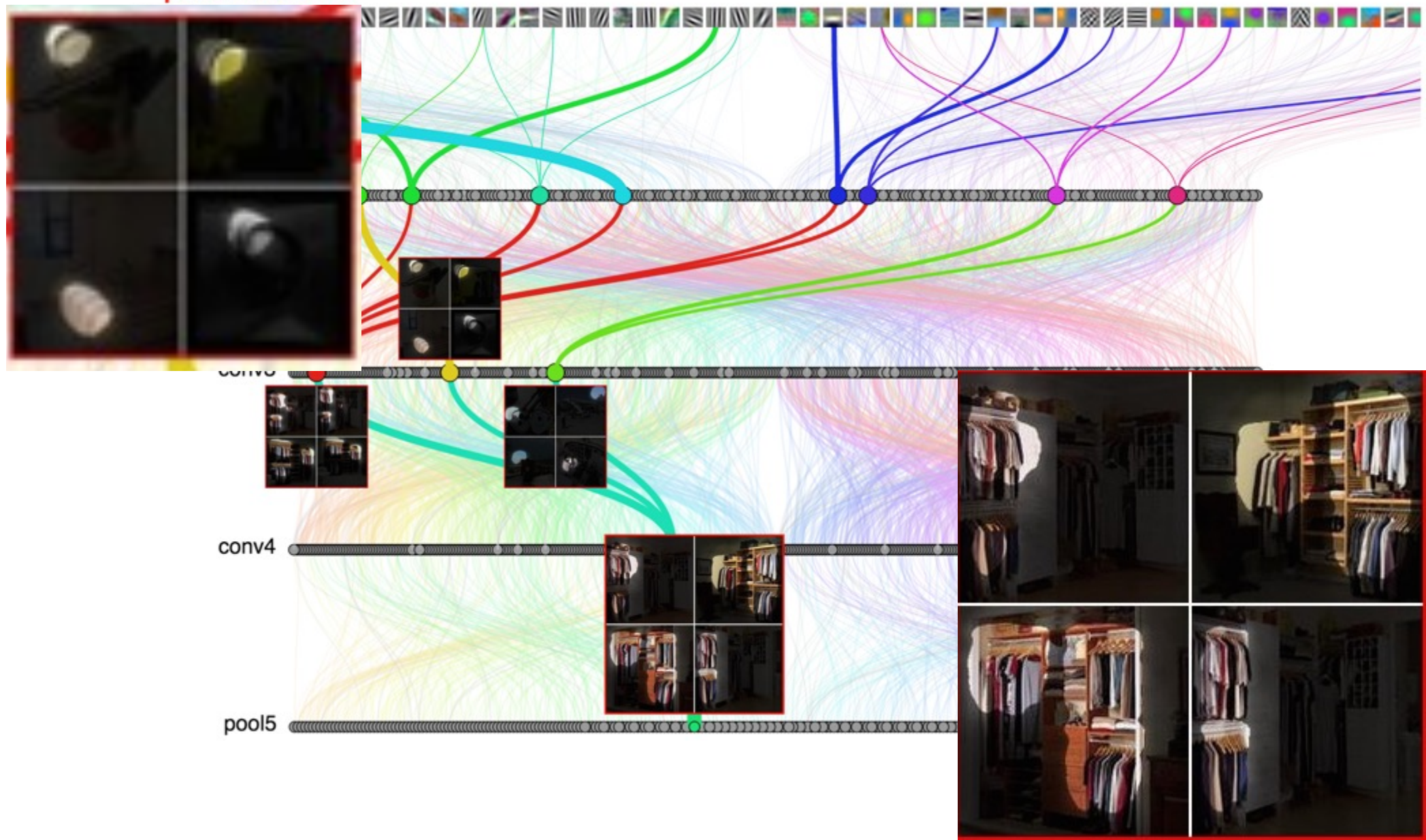
Looking inside the network:

DrawNet: (search: drawnet MIT)

<http://people.csail.mit.edu/torralba/research/drawCNN/drawNet.html>

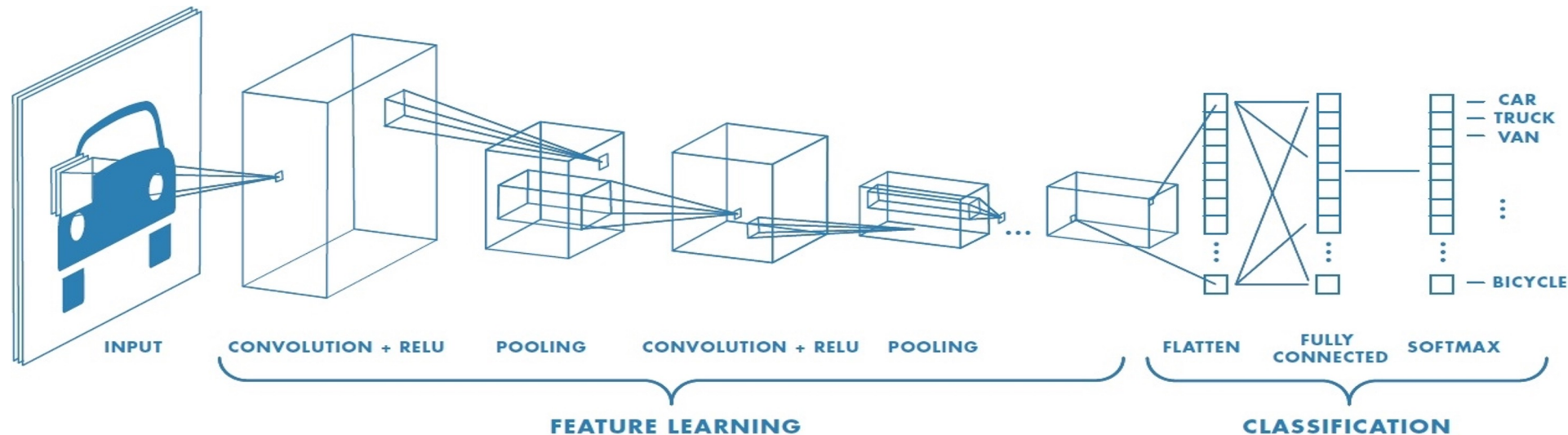


placesCNN



Parameters of a model

- All the weights and biases (billions!)
- Hyper-parameters (training) – learning rate, number of epochs, etc..
- Hyper-parameters (testing) – any data pre-processing



Testing a model

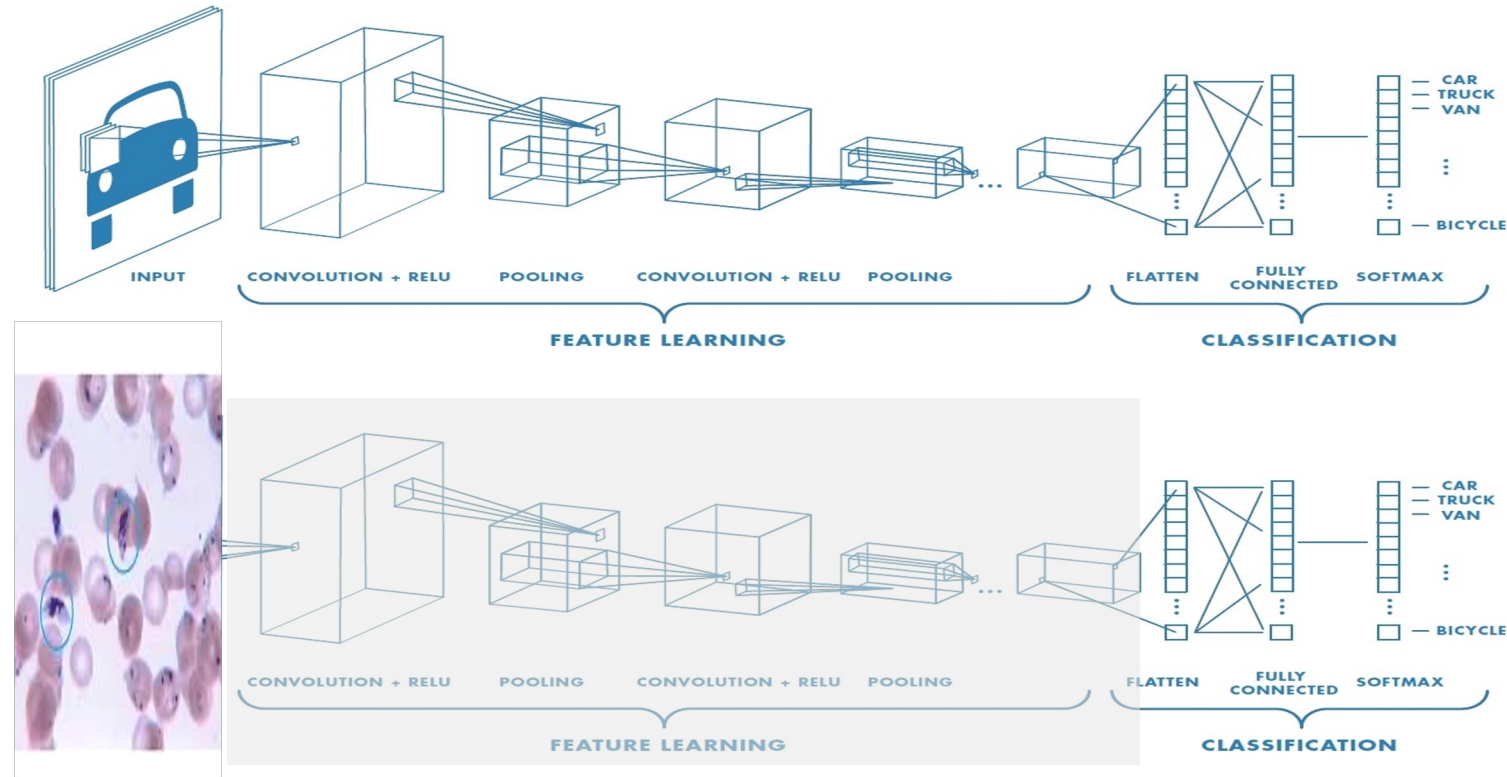
- Goal of the model is to be effective on *unseen* data.
- Machine learning datasets are broken up into training data, validation data and testing data.
- In best research paradigm, testing data is only ever seen once – validation data is used to select hyperparameters
- “Validity” of the machine learning model is how well it works on testing data.

- Imagenet
 - 14 million images
 - 1000 classes
- Now ~90% accuracy (top-5)



Fine-Tuning for Deep Learning

- Start with Imagenet
- Update those parameters with your own images and labels. Can often keep many of the weights the same, so much less training data is needed



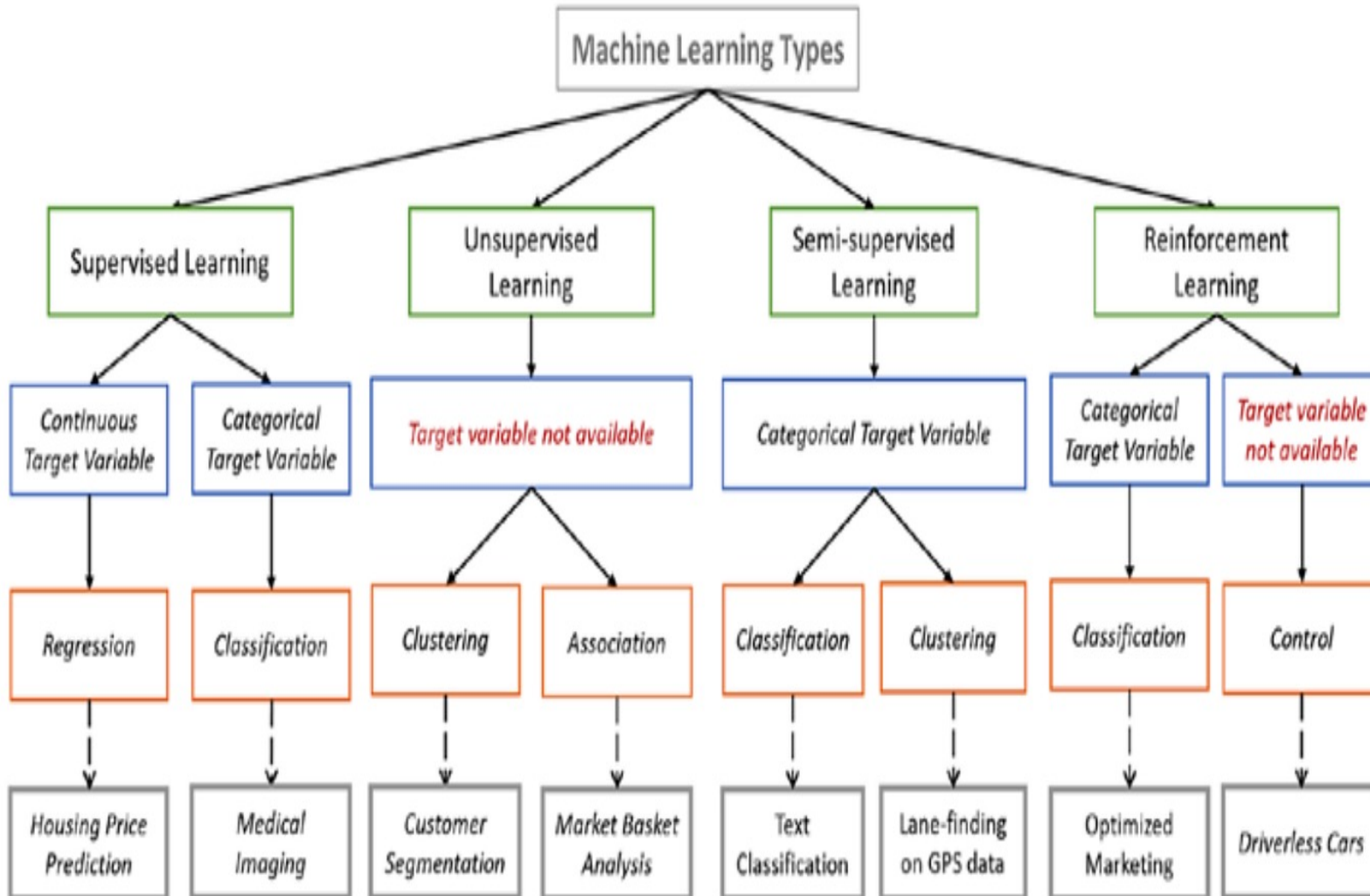
The “big” change in ML/AI recently is generative AI. Generative Models are like GPT:

“I woke up this morning feeling....”

“The sky was a beautiful shade of....”

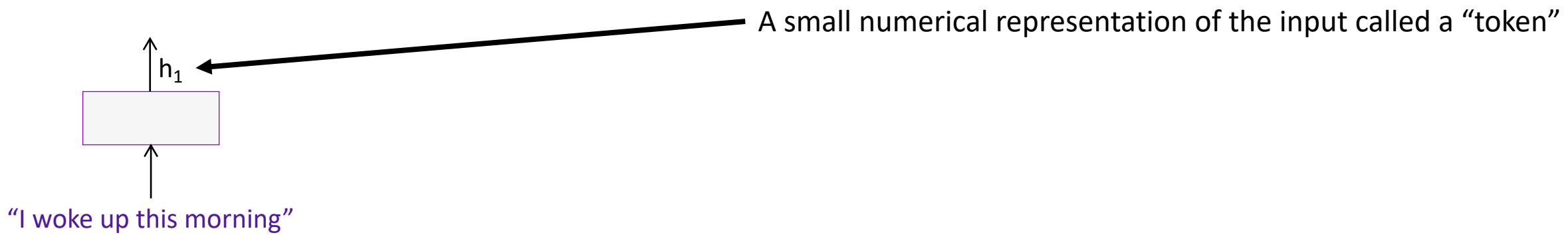
“My favorite food is....”

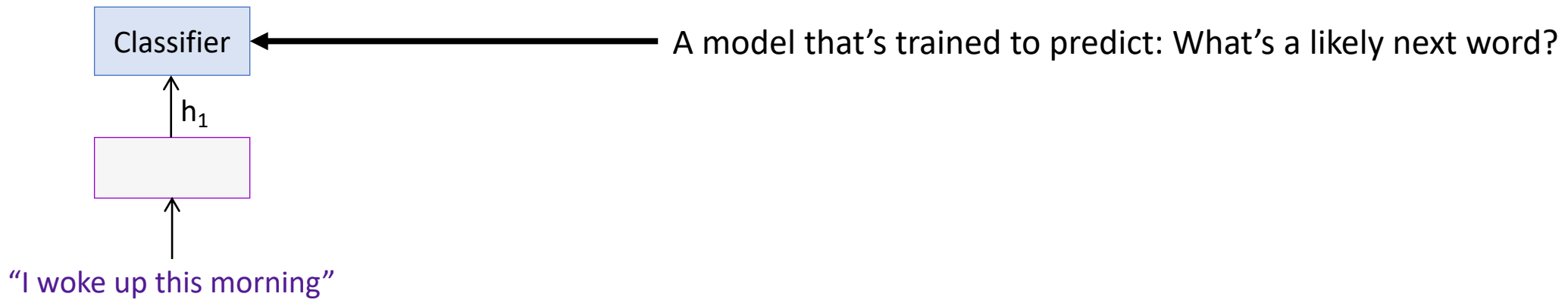
Very broad types of machine learning

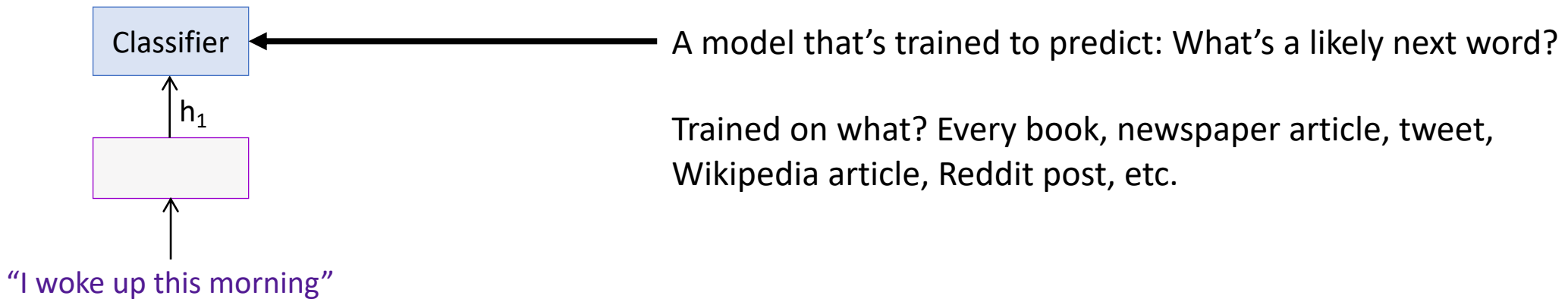


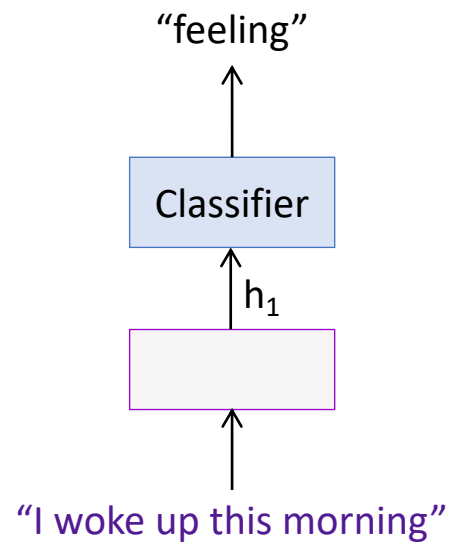
Generative AI

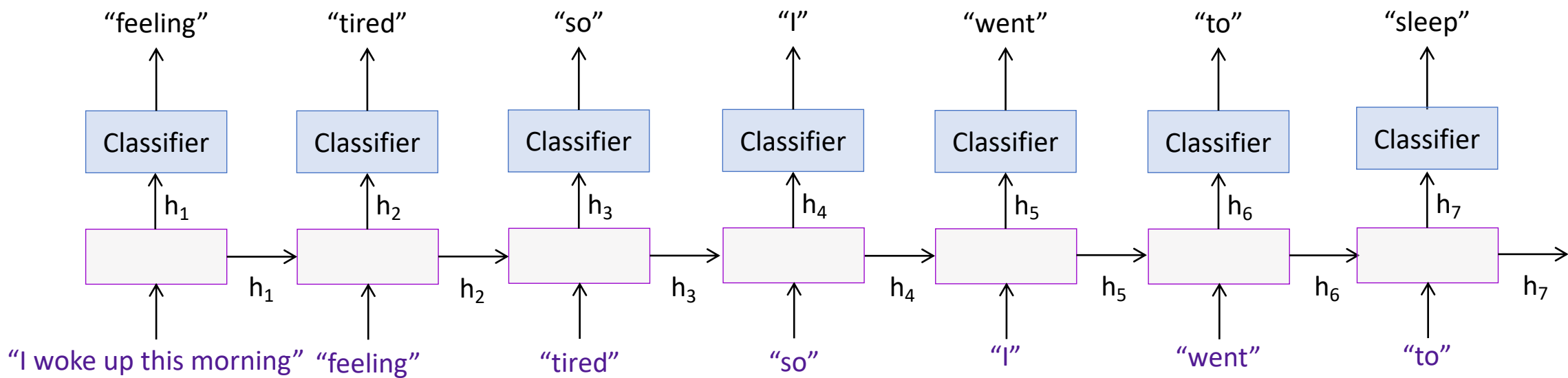
“I woke up this morning”

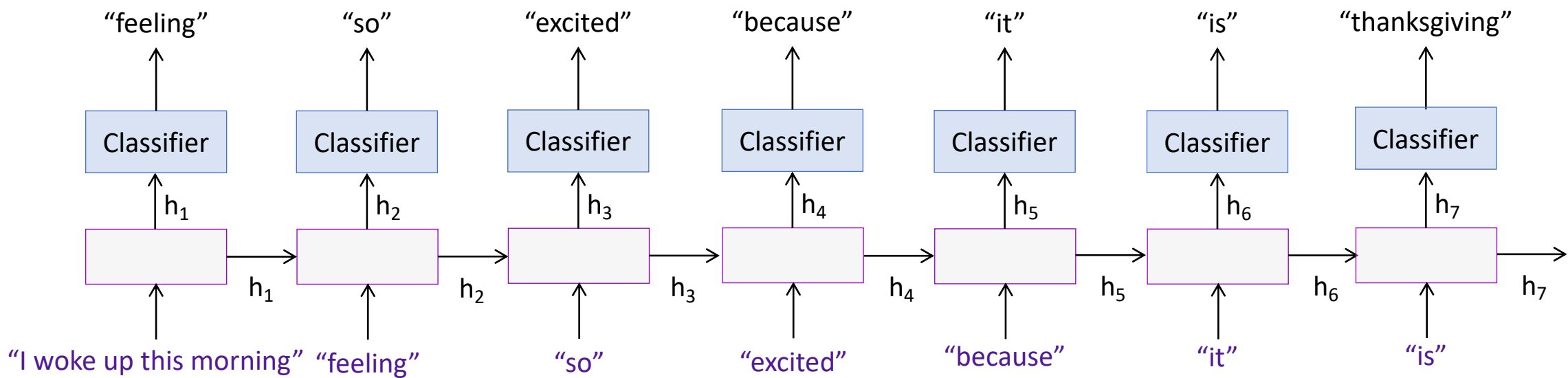


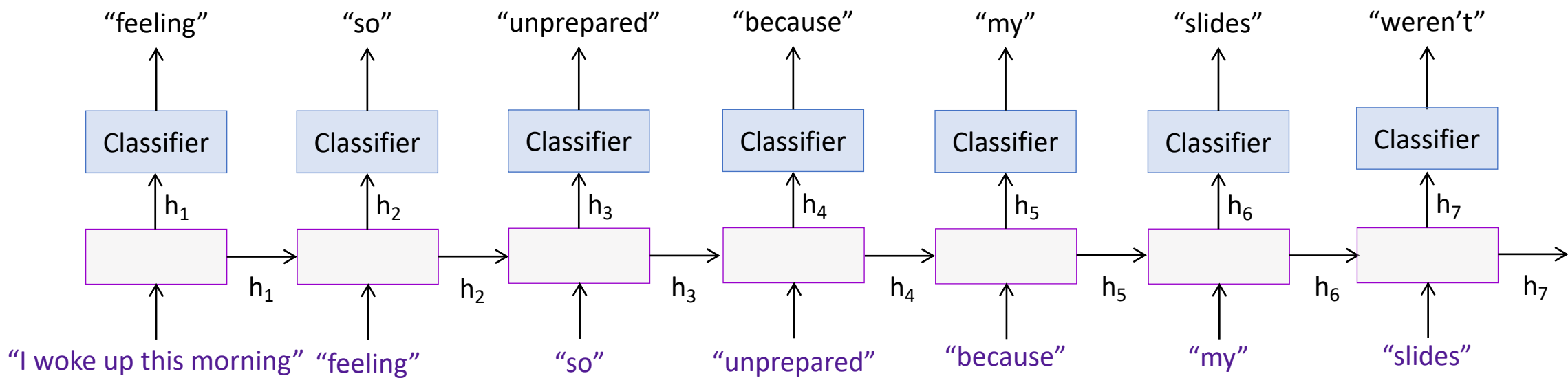












ChatGPT

- Modern tools put a lot of infrastructure around this “next word prediction”
- But the general idea holds – take all of the text that humans have ever created, and learn the probability of words and phrases occurring with each other.
- Given a new input, predict (based on the probabilities you’ve learned) what the most likely next phrase/response is.
- Improved by Reinforcement Learning: Make the prediction and then let a human provide feedback on how good it is. Helps to learn “better” answers, but *implemented* as predict the next word better, and sometimes “check to see if the answer is problematic and restart”.

20 Questions



Canned 20 questions:

ChatGPT 5.2 ▾

Can we play 20 questions? You pick a physical object and I'll try to ask questions

Yes. I've picked an object. Ask your first question.



do I like it most because it gives off light or because it is very fragile?



Because it gives off light.



is it a lightbulb?

Yes.



Stable Diffusion 2.1 Demo

Stable Diffusion 2.1 is the latest text-to-image model from StabilityAI. [Access Stable Diffusion 1 Space here](#)

For faster generation and API access you can try [DreamStudio Beta](#).

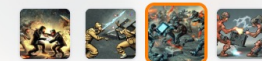
Not just text...

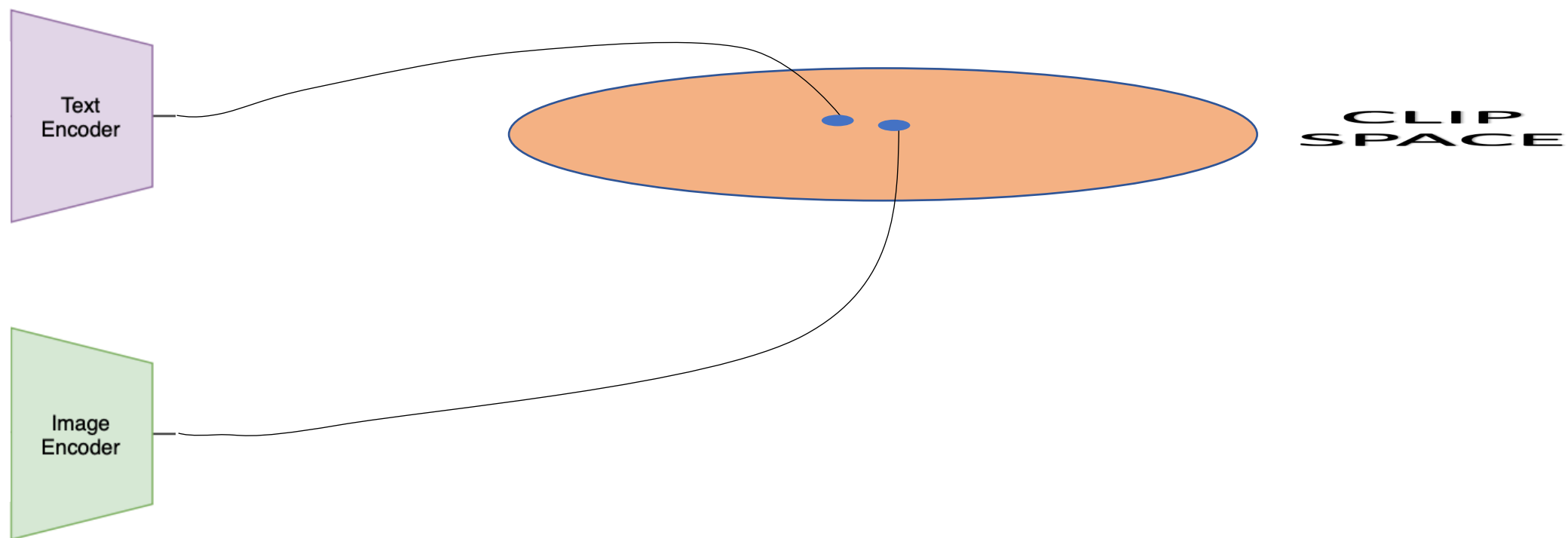
- Scrape all of the images from the internet and their associated text/captions
- Learn the relationship between text and images
- Train a model to create new images that are aligned with particular captions

A computer fighting in a war

Generate image

Enter a negative prompt

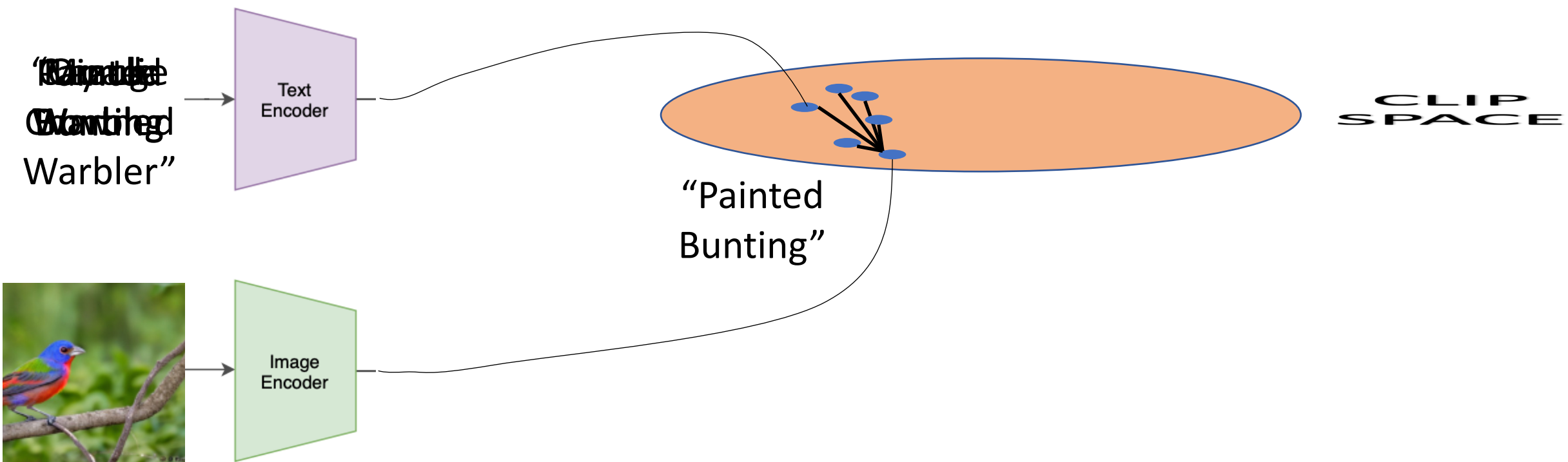




CLIP (Contrastive Language Image Pre-Training), is the technical basis that allows current approaches to link images and text.

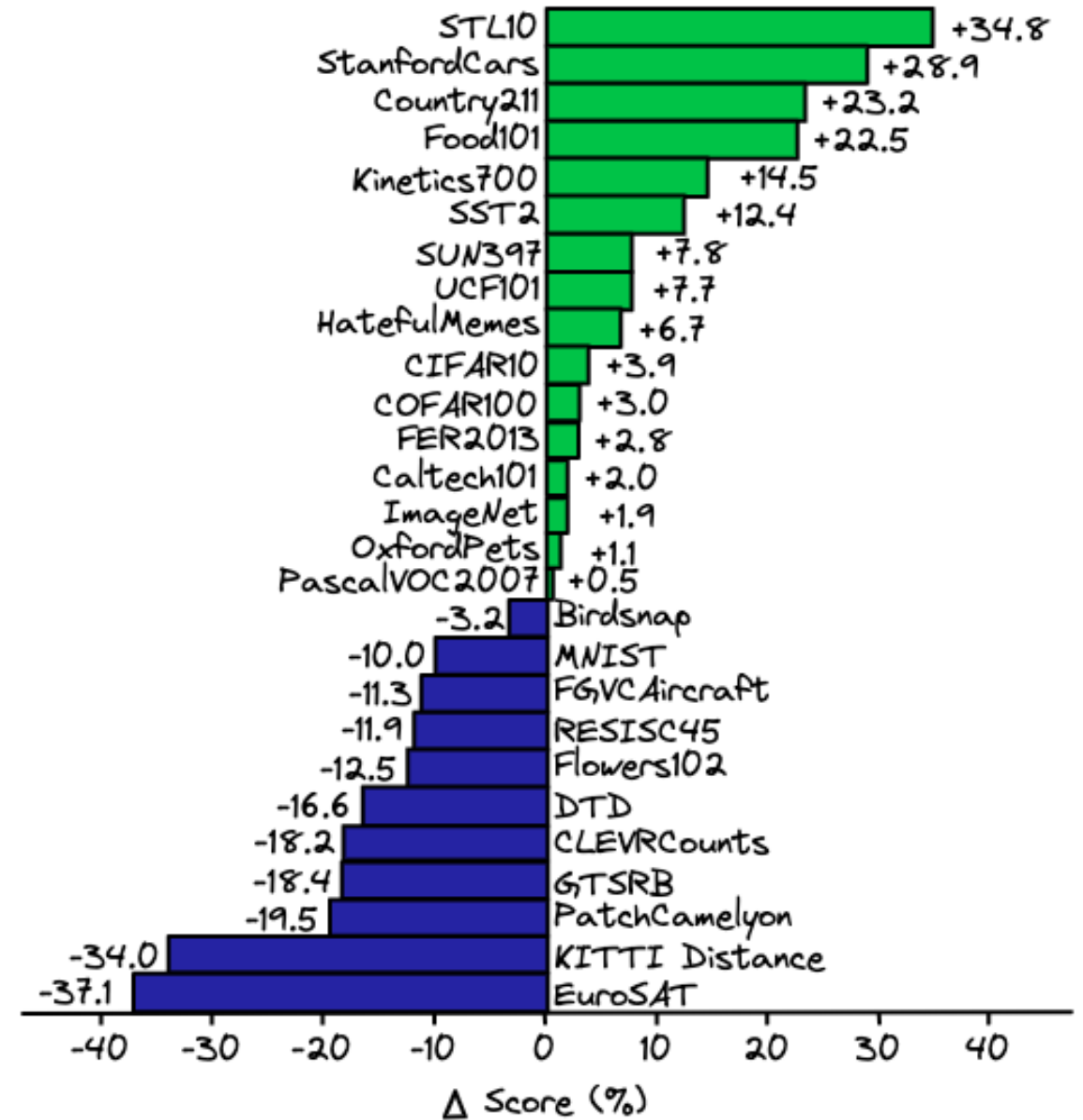
Trained on 5 billion pairs of (image, caption), CLIP has learned to embed images and related text to nearby locations (in a high-dimensional vector space).

This is surprisingly powerful.



This is surprisingly powerful. It also makes very quick, easy image classifiers

CLIP based classifiers outperform hand-tuned, specially trained classifiers for many, many datasets



If you have a problem for which you need image classification, you no longer need to talk to Computer Vision faculty. Now any good undergrad can make you a classifier for exactly the classes you care about.

This is surprisingly powerful. It also makes very quick, easy image classifiers

The Unreasonable Effectiveness of Data

Alon Halevy, Peter Norvig, and Fernando Pereira, *Google*

... even when your data is much less clean
that you'd hope...



Blue Beach
Umbrellas, Point
Of Rocks, Crescent
Beach, Siesta Key -
Spiral Notebook



BMW-M2-M-
Performance-Dekor-
Long-Beach-Blue-05



Becoming More Than a
Good Bible Study Girl:
Living the Faith after Bible
Class Is Over [...]



"Dynabrade 52632
4-1/2"" Dia. Right
Angle Depressed
Center Wheel
Grinder (Replaces
50306 and 50346)"

Example images + captions from the CLIP training data



Peugeot 308 2013
sedan



Episcopia Ortodoxa
a Maramuresului si
Satmarului are un
nou Arhiereu vicar



DON QUIJOTE
DE LA MANCHA
(SELECCIÓN DE
TEXTOS)

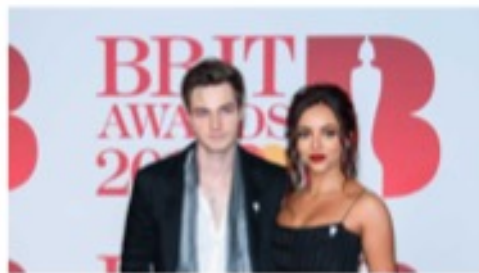


Żeński i męski
portret Dama
outdoors i facet
Ślubna para [...]

Example images + captions from the CLIP training data



18fcd9e025205
Fila Omnispeed
Men Us 10 Multi
Color Running
Shoe in Blue for
Men - Lyst



Little Mix's Jade
Thirlwall has 'split'
from her boyfriend
Jed Elliot



Saarinen Style:
M70 Womb
Ottoman Mcm
Furniture, Selling
Furniture, [...]



Europe, Italy

Example images + captions from the CLIP training data



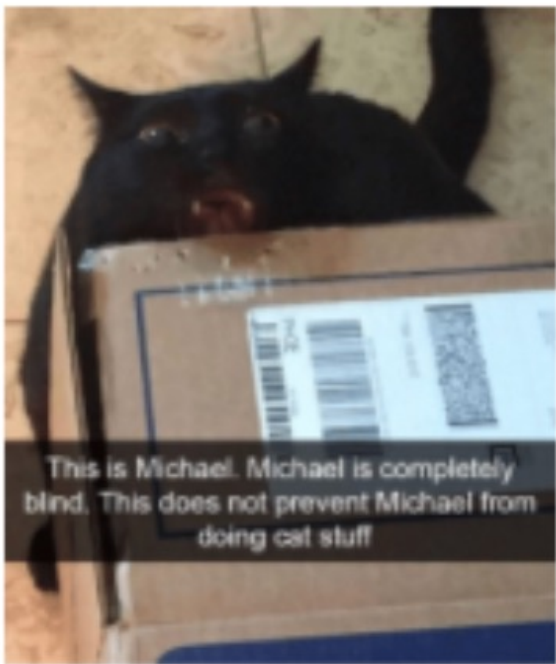
Mouser painting by
Lynda Nolte



"Oil on Canvas -
""Dignan"" - Dignan,
the loner, w...



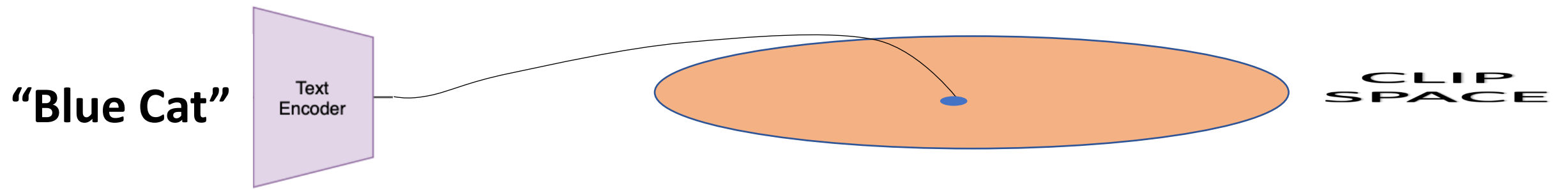
Drawings On Black
Paper Ideas ; Drawings
On Black ...



You bite that box Michael

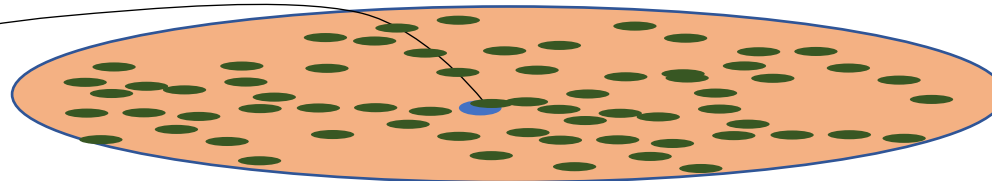
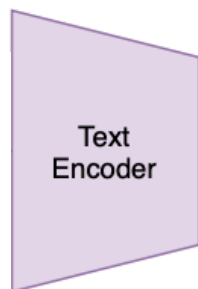
Michael, Stuff, and
Cat: This is Michael.
Michael ...

Example images + captions from the CLIP training data



Application 1: Query a large image dataset with arbitrary English

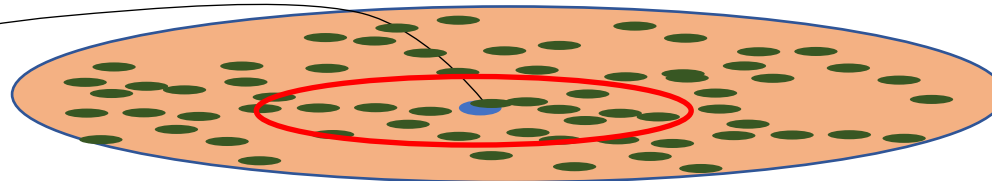
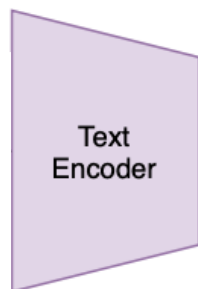
“Blue Cat”



**CLIP
SPACE**

**CLIP Features of all images in
your database**

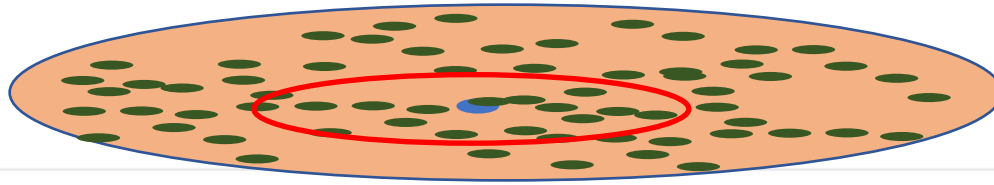
“Blue Cat”



**CLIP
SPACE**

**CLIP Features of all images in
your database**

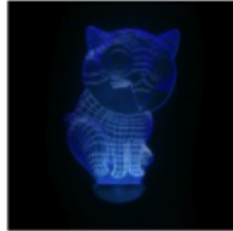
**Retrieve nearby images as
search results**



blue cat



Russian Blue Postcards (Package of 8)



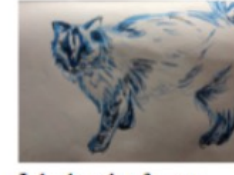
CAT, LED Lamps, slingly, FamilyTrophy.com - Family...



Halloween, Cat, Paper Plate, Toddler, Preschool



Blue Wooden Cat



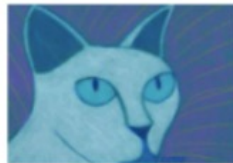
Ink sketch of a cat



Blue Eyes Greeting Card



Sitting cat low poly in Gloss Blue Porcelain



Tonkinese cat drawing BZTAT



Blue Cat Full Moon Postcard



Kitten Digital Art - Soft Kitty by William Paul M...

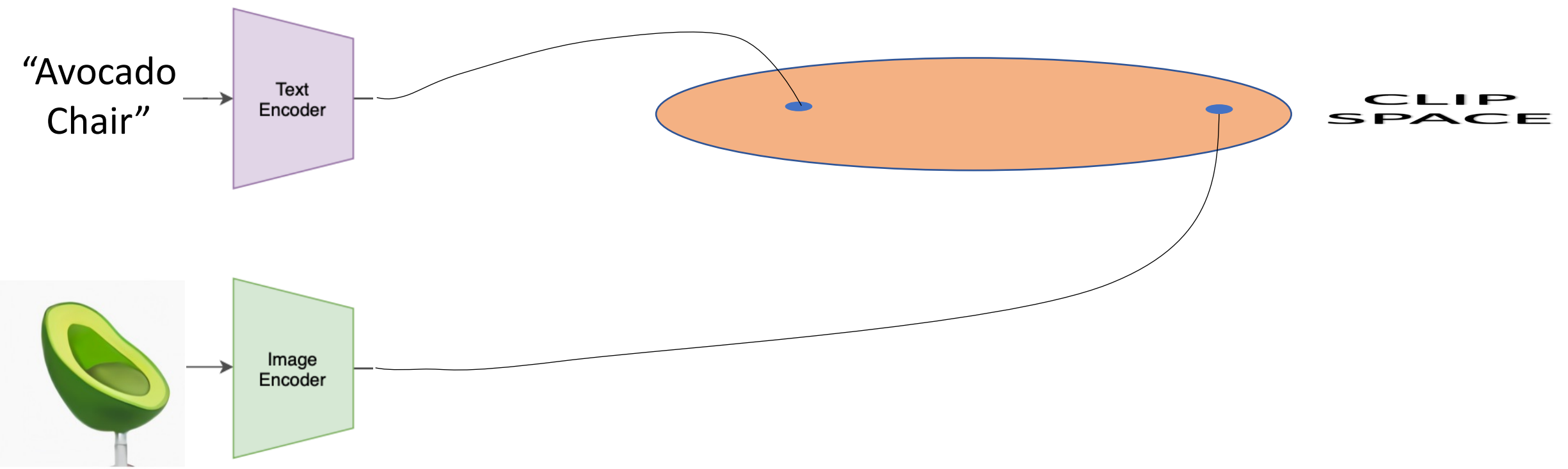


Blue and Black Panther Football Colors Acrylic Pai...



Tonkinese drawing - nicture #10

This allows generalizable image search for any feature you can name, even if that was not planned for when data was captured or stored.



Application 2: Dall-E. ("Generating images from Text")

- Your text is mapped into CLIP space to create an image target.
- Then a random starting image is created and mapped into CLIP space.
- Then gradient descent is used to update pixel values to change the image --- so that the CLIP embedding moves closer to the target

Shockingly Good

Prompt: Luke Skywalker ordering a burger and fries from the Death Star canteen.



Shockingly Good

Prompt: Black adventure van, geared out, apocalyptic, dystopian van, realistic, roof rack, in dark abandoned neighborhood, van geared out



Shockingly Good (*sometimes*)

Why would this happen?

The CLIP model mapping text to images is trained with images and captions --- of all possible objects, what is most likely to be in an image with a caption that says “no elephant”?

Midjourney Bot ✓ BOT Today at 4:04 AM
An empty room with no elephants. Not a single elephant in sight. --v 6.0 - @Gedas (fast)



“Multimodal AI”



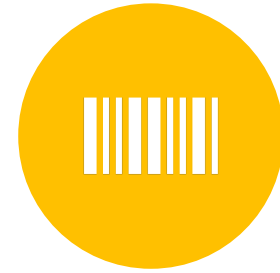
TEXT



IMAGE



AUDIO/SOUND



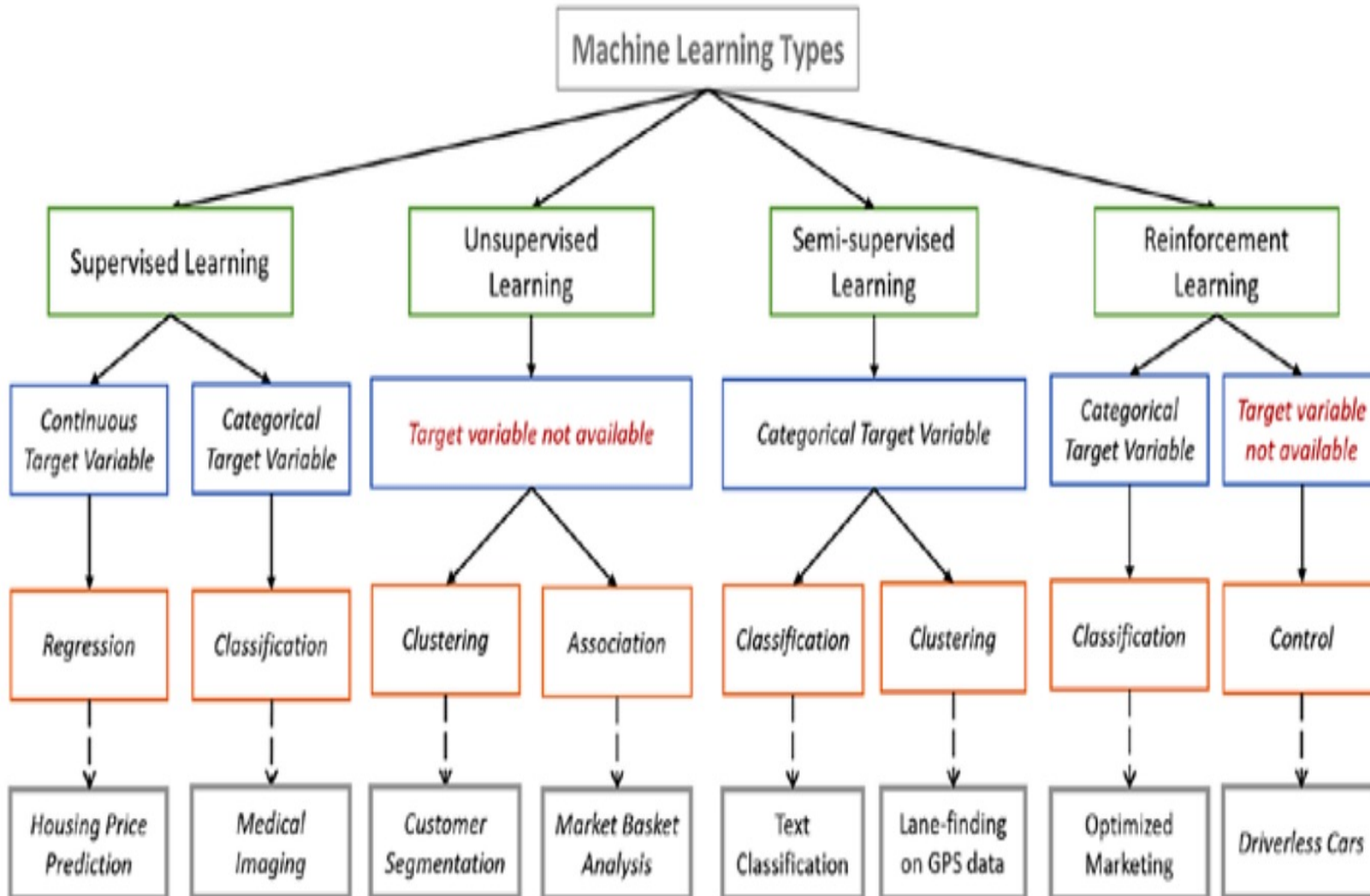
CODES

The approach of “tokenizing” inputs, and creating unified embeddings across modalities generalizes beyond text and images, also includes audio, 3D cameras, video data etc...

Last thoughts on AI

1. Generative AI offers much much easier approaches for computation to directly touch the real world
2. Works extremely well if you stay within the distributions for which it was trained (next word! Images and captions...), but don't ask it to write a long document...
3. Approaches to create guardrails, (e.g. so that Dall-E doesn't create inappropriate images) are kludgy and incomplete

Very broad types of machine learning



Generative AI

Agentic AI

Generative AI = Agentic AI if you connect words to actions

Every computer has a text window where you type a sentence and the machine does it: the terminal.

```
$ ls ~/Documents
```

```
Lectures      Grants      Reviews      taxes2025.pdf
```

```
$ wc -w thesis_draft.txt
```

```
41,752 thesis_draft.txt
```

What one line of text can do

```
$ say "good morning, class"
```

Your laptop speaks, out loud.

```
$ mv ~/Downloads/*.pdf ~/receipts/
```

File a PDF somewhere

```
$ git push
```

Push changes I made to my copy of the website to the internet

```
$ curl api.twilio.com/.../Messages -d ...
```

Send a text message to someone.

```
$ for f in *.jpg; do sips -Z 800 $f; done
```

Resize 1000 image files.

One sentence each. No programming. Each one reaches further out of the computer than the last.

What one line of text can also do

```
$ curl api.stripe.com/v1/charges -d amount=50000
```

Charges \$500 to a credit card.

```
$ crontab: 0 7 * * * python report.py
```

Runs every morning at 7:00

```
$ ssh lab-server 'shutdown now'
```

Turns off a computer in another building.

```
$ rm -rf ~/Documents
```

Delete all the documents in the Documents folder and all subfolders.